$S°(298.15\ K) = 186.251 \pm 0.04\ J \cdot K^{-1} \cdot mol^{-1}$

$\Delta_f H°(0\ K) = -66.911 \pm 0.34\ kJ \cdot mol^{-1}$
$\Delta_f H°(298.15\ K) = -74.873 \pm 0.34\ kJ \cdot mol^{-1}$

Vibrational Frequencies and Degeneracies ν, cm^{-1}
2916.5 (1)
1534.0 (2)
3018.7 (3)
1306 (3)

Ground State Quantum Weight: [1] $\sigma = 12$
Point Group: T_d
Bond Distance: C–H = 1.091 ± 0.002 Å
Bond Angle: H–C–H = 109.4712°
Product of the Moments of Inertia: $I_A I_B I_C = 1.499 \times 10^{-115}\ g^3 \cdot cm^6$.

Enthalpy of Formation

Rossini[1] measured the enthalpy of combustion of methane gas. His value at 298.15 K was corrected to the presently accepted molecular weight of water. The enthalpy of formation was calculated using −68.3174 and −94.0540 kcal·mol^{-1} for the enthalpy of formation of H_2O(l) and CO_2(g), respectively.

Heat Capacity and Entropy

Stevenson and Ibers,[2] calculated the bond distance from an analysis of available spectroscopic data. Vibrational frequencies listed by Jones and McDowell,[3] are consistent with their results of a force constant calculation. The principal moments of inertia are: $I_A = I_B = I_C = 0.5313 \times 10^{-39}\ g \cdot cm^2$.

References

[1]F. D. Rossini, J. Res. Nat. Bur. Stand. 6, 37 (1931).
[2]D. P. Stevenson and J. A. Ibers, J. Chem. Phys. 33, 100 (1960).
[3]L. H. Jones and R. S. McDowell, J. Mol. Spectrosc. 3, 632 (1959).

Enthalpy Reference Temperature = T_r = 298.15 K Standard State Pressure = $p°$ = 0.1 MPa

	J·K^{-1}mol^{-1}			kJ·mol^{-1}			
T/K	$C_p°$	$S°$	$-[G°-H°(T_r)]/T$	$H°-H°(T_r)$	$\Delta_f H°$	$\Delta_f G°$	log K_f
0	0.	0.	INFINITE	−10.024	−66.911	−66.911	INFINITE
100	33.258	149.500	216.485	−6.698	−69.644	−64.353	33.615
200	33.473	172.577	189.418	−3.368	−72.027	−58.161	15.190
250	34.216	180.113	186.829	−1.679	−73.426	−54.536	11.395
298.15	35.639	186.251	186.251	0.	−74.873	−50.768	8.894
300	35.708	186.472	186.252	0.066	−74.929	−50.618	8.813
350	37.874	192.131	186.694	1.903	−76.461	−46.445	6.932
400	40.500	197.356	187.704	3.861	−77.969	−42.054	5.492
450	43.374	202.291	189.053	5.957	−79.422	−37.476	4.350
500	46.342	207.014	190.614	8.200	−80.802	−32.741	3.420
600	52.227	215.987	194.103	13.130	−83.308	−22.887	1.993
700	57.794	224.461	197.840	18.635	−85.452	−12.643	0.943
800	62.932	232.518	201.675	24.675	−87.238	−2.115	0.138
900	67.601	240.205	205.532	31.205	−88.692	8.616	−0.500
1000	71.795	247.549	209.370	38.179	−89.849	19.492	−1.018
1100	75.529	254.570	213.162	45.549	−90.750	30.472	−1.447
1200	78.833	261.287	216.895	53.270	−91.437	41.524	−1.807
1300	81.744	267.714	220.558	61.302	−91.945	52.626	−2.115
1400	84.305	273.868	224.148	69.608	−92.308	63.761	−2.379
1500	86.556	279.763	227.660	78.153	−92.553	74.918	−2.609
1600	88.537	285.413	231.095	86.910	−92.703	86.088	−2.810
1700	90.283	290.834	234.450	95.853	−92.780	97.265	−2.989
1800	91.824	296.039	237.728	104.960	−92.797	108.445	−3.147
1900	93.188	301.041	240.930	114.212	−92.770	119.624	−3.289
2000	94.399	305.853	244.057	123.592	−92.709	130.802	−3.416
2100	95.477	310.485	247.110	133.087	−92.624	141.975	−3.531
2200	96.439	314.949	250.093	142.684	−92.521	153.144	−3.636
2300	97.301	319.255	253.007	152.371	−92.409	164.308	−3.732
2400	98.075	323.413	255.854	162.141	−92.291	175.467	−3.819
2500	98.772	327.431	258.638	171.984	−92.174	186.622	−3.899
2600	99.401	331.317	261.359	181.893	−92.060	197.771	−3.973
2700	99.971	335.080	264.020	191.862	−91.954	208.916	−4.042
2800	100.489	338.725	266.623	201.885	−91.857	220.058	−4.105
2900	100.960	342.260	269.171	211.958	−91.773	231.196	−4.164
3000	101.389	345.690	271.664	222.076	−91.705	242.332	−4.219
3100	101.782	349.021	274.106	232.235	−91.653	253.465	−4.271
3200	102.143	352.258	276.498	242.431	−91.621	264.598	−4.319
3300	102.474	355.406	278.842	252.662	−91.609	275.730	−4.364
3400	102.778	358.470	281.139	262.925	−91.619	286.861	−4.407
3500	103.060	361.453	283.391	273.217	−91.654	297.993	−4.447
3600	103.319	364.360	285.600	283.536	−91.713	309.127	−4.485
3700	103.560	367.194	287.767	293.881	−91.798	320.262	−4.521
3800	103.783	369.959	289.894	304.248	−91.911	331.401	−4.555
3900	103.990	372.658	291.982	314.637	−92.051	342.542	−4.588
4000	104.183	375.293	294.032	325.045	−92.222	353.687	−4.619
4100	104.363	377.868	296.045	335.473	−92.422	364.838	−4.648
4200	104.531	380.385	298.023	345.918	−92.652	375.993	−4.676
4300	104.688	382.846	299.967	356.379	−92.914	387.155	−4.703
4400	104.834	385.255	301.879	366.855	−93.208	398.322	−4.729
4500	104.972	387.612	303.758	377.345	−93.533	409.497	−4.753
4600	105.101	389.921	305.606	387.849	−93.891	420.679	−4.777
4700	105.223	392.182	307.424	398.365	−94.281	431.869	−4.800
4800	105.337	394.399	309.213	408.893	−94.702	443.069	−4.822
4900	105.445	396.572	310.973	419.432	−95.156	454.277	−4.843
5000	105.546	398.703	312.707	429.982	−95.641	465.495	−4.863
5100	105.642	400.794	314.414	440.541	−96.157	476.722	−4.883
5200	105.733	402.847	316.095	451.110	−96.703	487.961	−4.902
5300	105.818	404.861	317.750	461.688	−97.278	499.210	−4.920
5400	105.899	406.840	319.382	472.274	−97.882	510.470	−4.938
5500	105.976	408 784	320.990	482.867	−98.513	521.741	−4.955
5600	106.049	410.694	322.575	493.469	−99.170	533.025	−4.972
5700	106.118	412.572	324.137	504.077	−99.852	544.320	−4.988
5800	106.184	414.418	325.678	514.692	−100.557	555.628	−5.004
5900	106.247	416.234	327.197	525.314	−101.284	566.946	−5.019
6000	106.306	418.020	328.696	535.942	−102.032	578.279	−5.034

PREVIOUS. March 1961 (1 atm) CURRENT: March 1961 (1 bar)